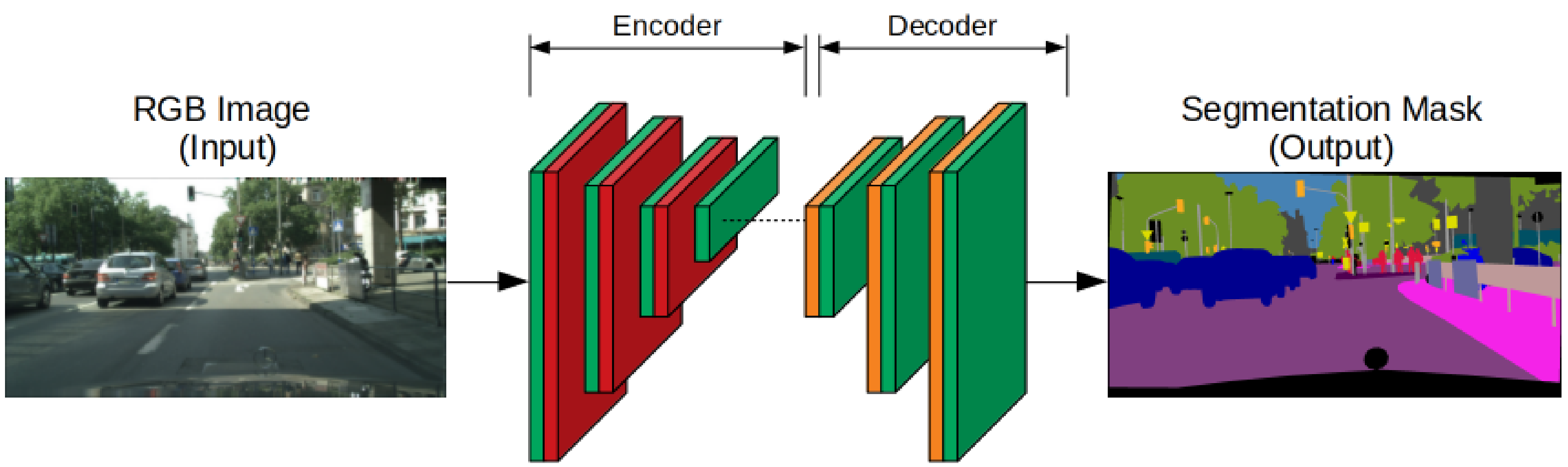
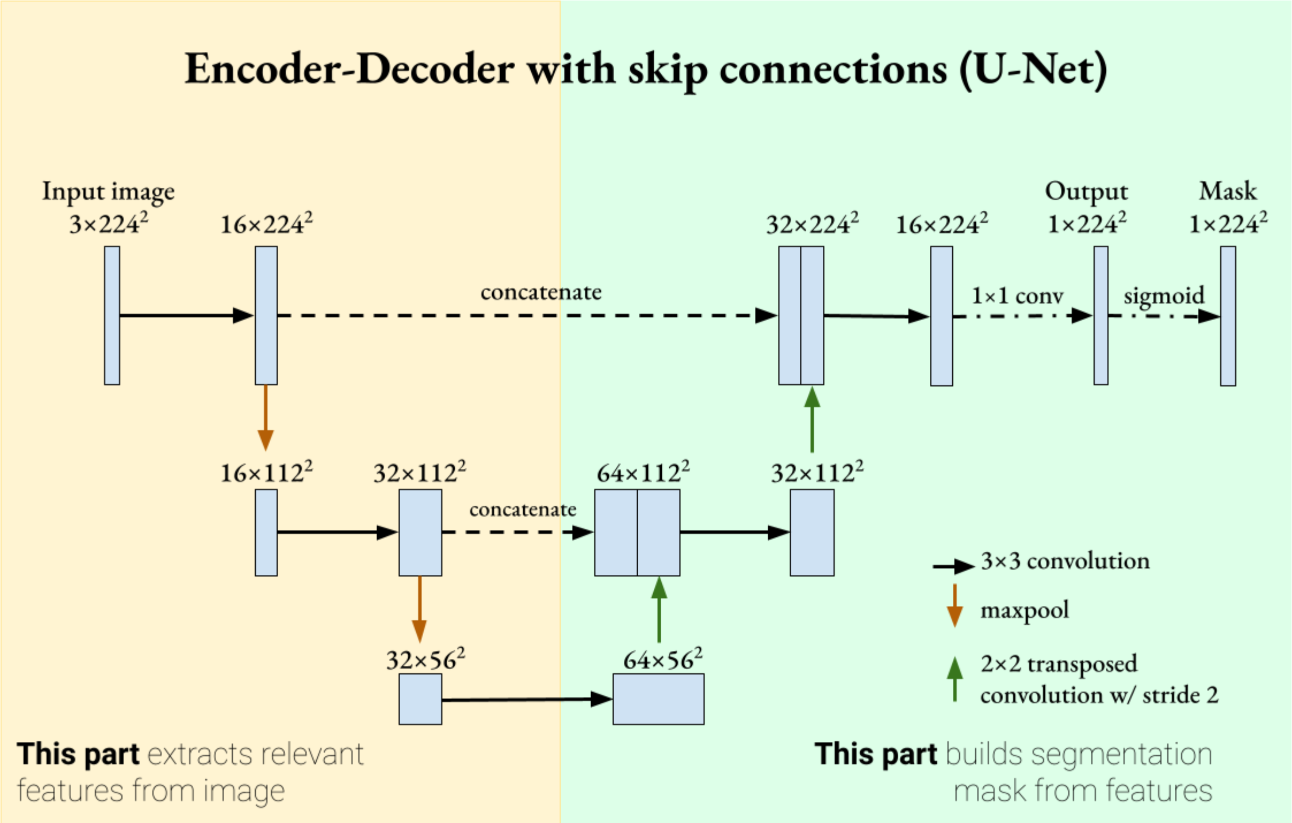
Semantic Segmentation: Every Pixel Counts



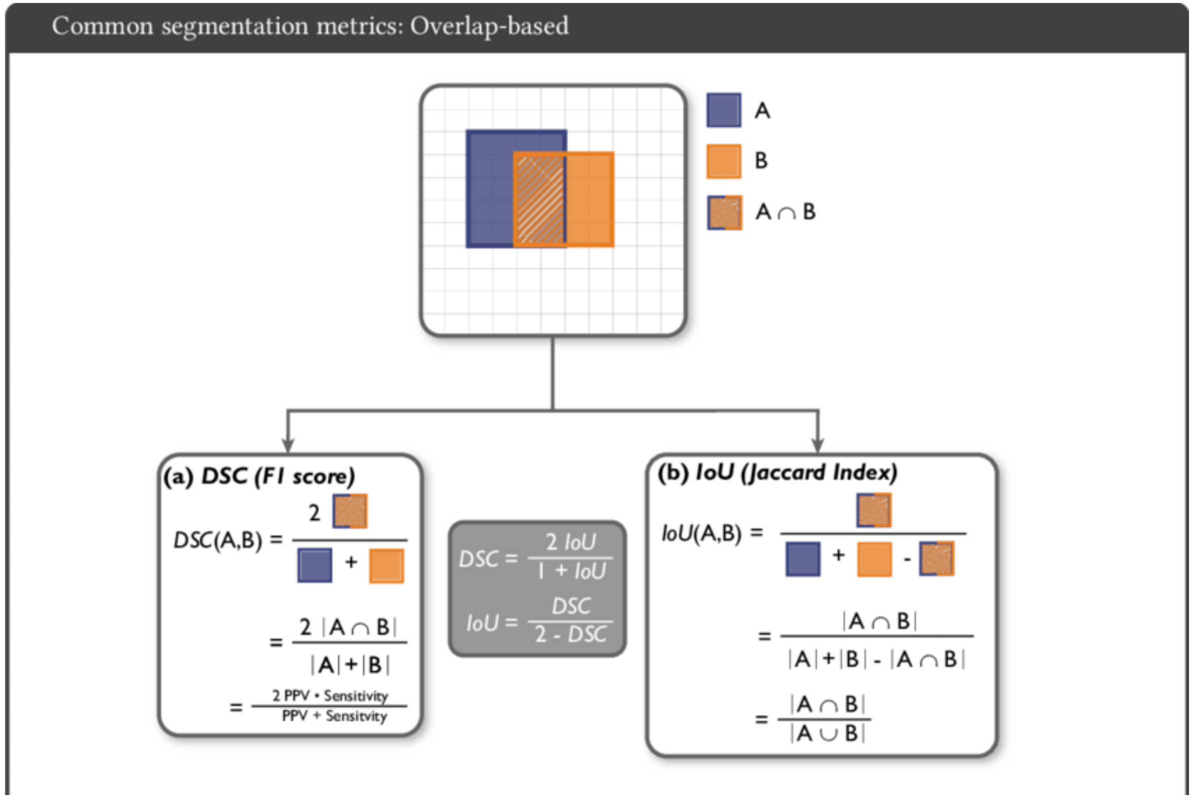
- Semantic Segmentation is the process of pixel-level classification, where every single pixel in an image is assigned a category label.
- The **U-Net** architecture is the gold standard for this task, particularly in medical and scientific fields.
- Unlike classification, the output of a segmentation model is a "mask"—a 2D grid of the same size as the input image.

Skip Connections: Recovering the Fine Details



- U-Net consists of a contracting path (Encoder) to capture context and a symmetric expanding path (Decoder) to enable precise localization.
- As images are downsampled in the encoder, fine spatial information is often lost.
- **Skip Connections** solve this by passing high-resolution feature maps from the encoder directly to the corresponding layers in the decoder.
- This allows the model to recover spatial detail, which is critical for creating high-precision masks in medical scans.

Dice Loss and Jaccard Index



- In segmentation, standard accuracy is often misleading because the background usually takes up 99% of the pixels.
- **Dice Loss**: Specifically designed to measure the overlap between the predicted mask and the ground truth.
- **Jaccard Index (IoU)**: Another precision metric used instead of standard accuracy to evaluate how well the predicted mask matches the target.